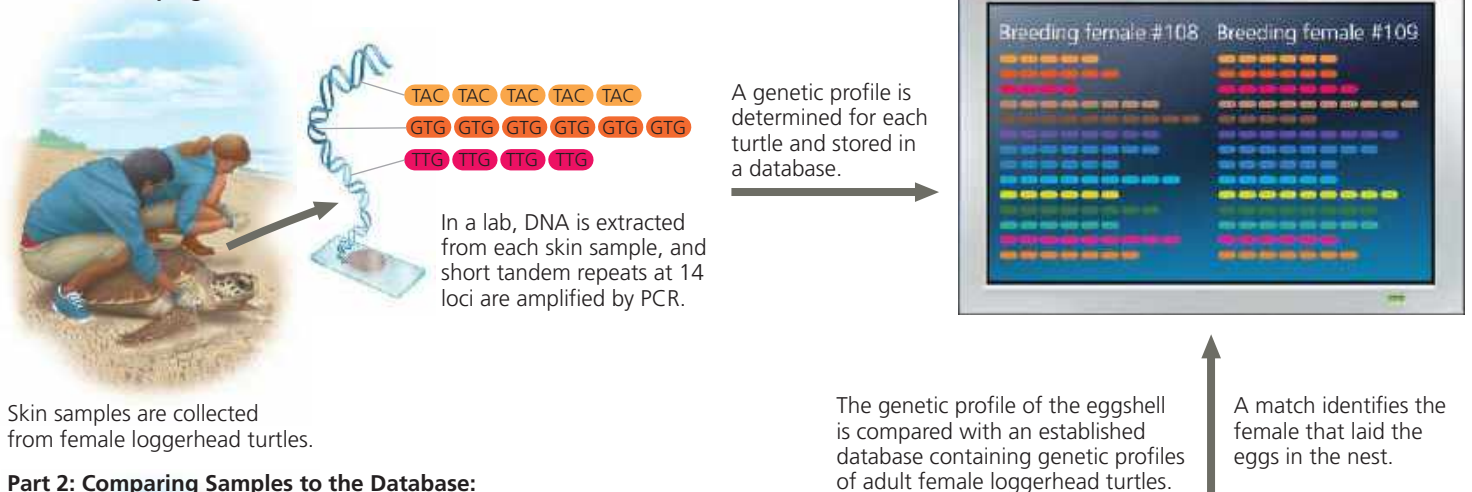
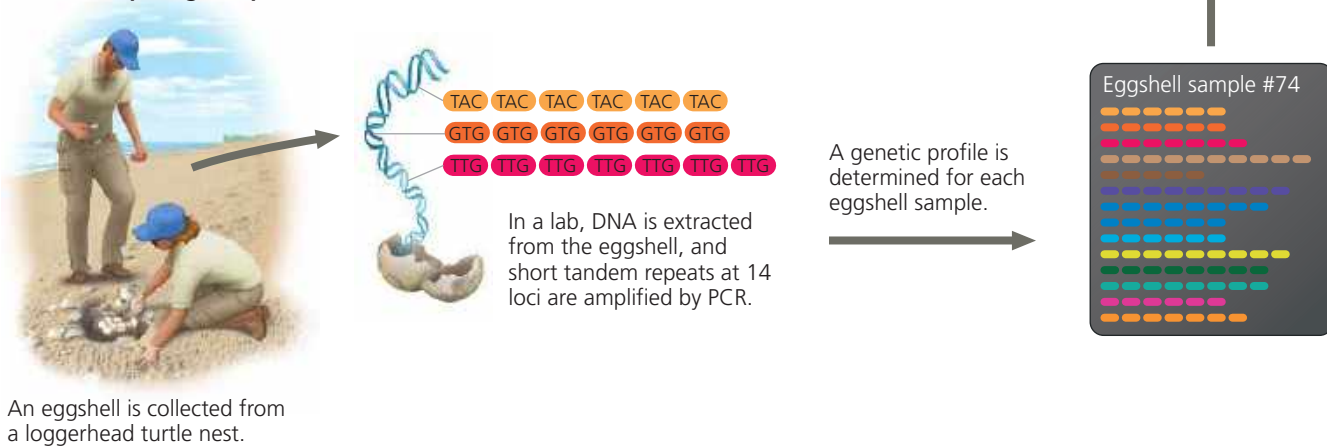


▼ **Figure 53.6** Using genetic profiles from loggerhead turtle eggshells to identify which female laid the eggs.

Part 1: Developing the Database:



Part 2: Comparing Samples to the Database:



VISUAL SKILLS Use the profiles displayed in the figure to determine which breeding female laid the eggs in the nest from which eggshell sample #74 was taken.

specific female. This approach allowed them to determine how many of the 198 females were breeding—and how many offspring each female produced—without having to disturb the females during egg laying.

Reproductive output for sexual organisms such as birds and mammals is typically measured as the average number of female offspring produced by the females in a given age-group. For some organisms, the number of offspring for each female can be counted directly; alternatively, molecular methods can be used (see Figure 53.6). Researchers directly counted the offspring of the Belding's ground squirrels, which begin to reproduce at age 1 year. The squirrels' reproductive output rises to a peak at 4–5 years of age and then gradually falls off in older females (see Table 53.1).

Age-specific reproductive rates vary considerably by species. Squirrels, for example, have a litter of two to six young once a year for less than a decade, whereas oak trees drop thousands of acorns each year for tens or hundreds of years.

Mussels and other invertebrates may release millions of eggs and sperm in a spawning cycle. However, a high reproductive rate will not lead to rapid population growth unless conditions are near ideal for the growth and survival of offspring, as you'll learn in the next section.

CONCEPT CHECK 53.1

- DRAW IT** Each female of a particular fish species produces millions of eggs per year. Draw and label the most likely survivorship curve for this species, and explain your choice.
- WHAT IF?** Suppose you are constructing a life table for a population of Belding's ground squirrels (see Table 53.1). If 485 individuals are alive at the start of year 0–1 and 218 are still alive at the start of year 1–2, what is the proportion alive at the start of each of these years (see column 3 in Table 53.1)?
- MAKE CONNECTIONS** A male stickleback fish attacks other males that invade its nesting territory (see Figure 51.2a). Predict the likely pattern of dispersion for male sticklebacks, and explain your reasoning.

For suggested answers, see Appendix A.